

AMPACITY	MVA RATING	COND TEMP	SHEATH TEMP	TOTAL LOSS	STATUS
312 A	8.13 MVA	90.0 degC	83.4 degC	19.855 W/m	Converged

1. Design Data & Input Parameters

Parameter	Formula / Symbol	Value	Unit
Conductor material		Cu	
Conductor size		120	mm2
Conductor type		stranded	
Conductor diameter Dc		13.6	mm
Conductor screen OD Dc_sc		14.6	mm
Max conductor temp theta_max		90	degC
Insulation material		XLPE	
Insulation screen OD Di		23.0	mm
Ins outer screen OD Di_sc		24.1	mm
Insulation thickness t_ins		4.50	mm
Relative permittivity er		2.5000	
Loss tangent tan(delta)		0.001000	
Screen type		Cu_wire	
Screen mean diameter ds		27.3	mm
Screen area As		35.0	mm2
Armour type		none	
Oversheath material		PVC	
Cable outer diameter De		31.5	mm
Rated voltage U		15.0	kV
U0 line-to-earth		8.700	kV
Frequency		60	Hz
Bonding		solid	
Formation		trefoil	
Depth to cable centre L		800	mm
Axial spacing s		31.5	mm
Ambient ground temp theta_a		35	degC
Soil thermal resistivity rho_soil		1.20	K.m/W

2. DC Resistance [IEC 60287-1-1 Cl.2.1.1]

Parameter	Formula / Symbol	Value	Unit
IEC 60228 R0 at 20 degC		1.5300e-04	ohm/m
Temperature coefficient a20		3.9300e-03	1/K
R0 at theta_amb = 35 degC		1.6202e-04	ohm/m
R0 at theta_max = 90 degC		1.9509e-04	ohm/m

3. AC Resistance at theta_max [IEC 60287-1-1 Cl.2.1.2, 2.1.4]

Parameter	Formula / Symbol	Value	Unit
ks strand constant		1.0000	
kp proximity constant		0.8000	
$xs4 = [ks(8\pi i f / R0) \times 10^{-7}]^2$		0.597463	
ys skin effect factor		0.003104	
$xp4 = [kp(8\pi i f / R0) \times 10^{-7}]^2$		0.382376	
yp" intermediate proximity		0.001988	
yp proximity factor		0.001638	
R at theta_max AC = $R0(1+ys+yp)$		1.9602e-04	ohm/m

4. Dielectric Losses [IEC 60287-1-1 Cl.2.2]

Parameter	Formula / Symbol	Value	Unit
Relative permittivity er		2.5000	
Loss tangent tan(delta)		0.001000	
U0 line-to-earth		8.7000	kV
Capacitance C = $er / (18 \ln(Di_sc / Dc_sc)) \times 10^{-9}$		0.275609	nF/m
$\omega = 2 \pi f$		376.9911	rad/s
$Wd = \omega C U0^2 \tan(\delta)$		7.8644e-03	W/m
TB880 GP7		Applied	

5. Screen / Sheath Setup [IEC 60287-1-1 Cl.2.3]

Parameter	Formula / Symbol	Value	Unit
Screen material		Cu	
Screen resistivity rho_s		1.7241e-08	ohm.m
Screen area As		35.00	mm2
Lay factor LFs		1.093845	
Screen Rs0 at 20 degC (with LFs)		5.3883e-04	ohm/m

6. Armour Losses [IEC 60287-1-1 Cl.2.4]

Parameter	Formula / Symbol	Value	Unit
Armour type		None	
lam2		0.000000	

7. Thermal Resistances [IEC 60287-2-1]

Parameter	Formula / Symbol	Value	Unit
rho_ins		3.50	K.m/W
rho_sc semiconducting screen		2.50	K.m/W
rho_tape bedding tape		6.00	K.m/W
T1_cs conductor screen		0.026555	K.m/W
T1_i insulation		0.253719	K.m/W
T1_si insulation screen		0.019281	K.m/W
T1_uwbt bedding tape under wire screen		0.043378	K.m/W
T1_raw = sum of layers		0.342933	K.m/W
T1 correction factor x1.00 (wire coverage check)		0.342933	K.m/W
T2 filler/bedding single-core = 0		0.000000	K.m/W
T3_raw oversheath		0.094634	K.m/W
T3 x1.6 touching trefoil factor		0.151415	K.m/W
u = 2L/De		50.8421	
T4 soil		2.287174	K.m/W
Sum T = T1+T2+T3+T4		2.781522	K.m/W

8. Sheath Losses at Convergence [IEC 60287-1-1 Cl.2.3]

Parameter	Formula / Symbol	Value	Unit
theta_s at convergence		83.4172	degC
Rs at theta_s		6.7312e-04	ohm/m
X sheath reactance		6.0481e-05	ohm/m
lam1' circulating current		0.027502	
lam1" eddy current		0.006631	
lam1 = lam1' + lam1"		0.034132	
lam2 armour		0.000000	

9. Ampacity Calculation [IEC 60287-1-1 Cl.1.4]

Parameter	Formula / Symbol	Value	Unit
Delta_theta = theta_max - theta_amb		55.00	K
Etop numerator		54.979474	
Ebot denominator		5.6154e-04	
I = sqrt(Etop/Ebot) exact		312.9042	A
I rounded TB880 GP2		312	A
MVA = sqrt(3) x U x I		8.129	MVA
Iterations		5	
Convergence		YES	

10. Temperature Profile Through Cable

Parameter	Formula / Symbol	Value	Unit
theta_c conductor target		90.0	degC
theta_c back-calculated		90.0000	degC
theta_s sheath surface		83.4172	degC
theta_j cable outer surface		80.4109	degC
theta_a ambient ground		35.0	degC

11. Loss Budget

Parameter	Formula / Symbol	Value	Unit
Conductor Wc = I ² x R		19.191675	W/m
Sheath Ws = lam1 x Wc		0.655057	W/m
Armour Wa = lam2 x Wc		0.000000	W/m
Dielectric Wd		7.8644e-03	W/m
Total		19.854596	W/m

12. Electrical Parameters

Parameter	Formula / Symbol	Value	Unit
Inductance L		0.166913	uH/m
Reactance X = omega L		62.9246	uOhm/m
Pos-seq resistance R1		202.7058	uOhm/m
Pos-seq impedance Z1		212.2478	uOhm/m
Capacitance C		0.275609	nF/m
Charging current Ic		0.9039	mA/m



Standards: IEC 60287-1-1:2023 | IEC 60287-2-1:2015 | CIGRE TB 880 guidance applied.
GP2: ampacity rounded down. GP6: eddy losses always calculated. GP7: dielectric losses always calculated. GP8: exact T4 formula. T3: 1.6x factor for touching trefoil.
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